

is possible using an arbitrary waveform generator (AWG). Modern AWGs can be programmed to directly generate RFID signals across HF and UHF bands and thus be used to simulate a variety of signals such as multiple reader or multiple tag signals with just one instrument. This reduces the time and cost of having to configure multiple signal generators.

The analysis device often needs quite deep memory to capture all of these lengthy interactions. Typically, the tag reader tries multiple queries and may command the tags to reduce their link frequency to verify that the tags are vacating the channel as required in some implementations. Real-time spectrum analysers (RTSAs) are capable of analysing these types of transactions.

An RTSA can directly verify the 60-second transmission and the 10-second silent period of ISO18000-7, with memory depth of more than 100 seconds in this application. This allows a thorough analysis of error conditions.

It is also possible to use multiple acquisitions to analyse hopping and bursting RFID signals. In this mode, the RTSA is set up to capture data for a user-defined period of time whenever a hop, and the associated trigger, occurs. This, in combination with a very high frame rate (over 48,000 fps), allows capture, analysis and demodulation of a hopping RFID signal.

Once the signal is captured, it can be analysed in a way that helps engineers to understand whether the reader and tag are performing as expected in the current RF environment, and if not, why. Measurements of

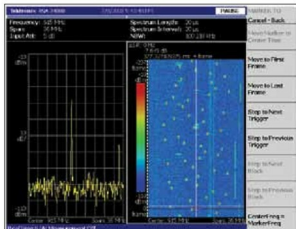


Fig. 4. Timing measurements of a 37ms hopping/burst signal, using markers in the spectrogram display (on the right), with an RTSA

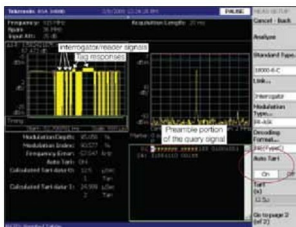


Fig. 5. Auto data rate select when testing ISO18000-6C (EPC GEN2). Note the multiple-yellow 'P' indicates the preamble. A yellow 'S' is used to indicate frame synch when it is visible



Fig. 6. A DPX display can show complex interactions between a tag reader and tags in the presence of many types of interference. This display shows that tags are responding when there is no interfering signal

bit time, CW time, and response time between the reader and the tag (known as turnaround time) give important insight into the reader and tag interaction and throughput. Checking amplitude glitches against frequency events can help spot root causes of errors. If a particular bit is not decoding properly, is it because of an error in the FSK or ASK modulation? Correlating data in various domains helps to answer these types of questions.

Latest RTSAs can correlate data in frequency, time, symbol and other domains, allowing extensive and quick analysis of complicated RF environments and physical-layer interactions. For ISO18000-6C (EPC GEN2) signals, which automatically change their data rate, these instruments can automatically detect the symbol rate and highlight the preamble, making analysis even easier.

Monitoring RFID co-channel interference

RFID transceivers must comply with local regulations regarding creating interference as well as be designed for optimal immunity to interference. Allocated spectrum, for instance, varies from 2 MHz in Singapore and Europe to 26 MHz in North America. This makes different modulation schemes and collision avoidance techniques favoured in different parts of the world.

Two approaches taken to implement collision avoidance and reduce self-interference are frequency hopping and listen-before-talk (LBT)/synchronisation of RFID readers. Frequency hopping is utilised in the United States according to FCC 47 CFR Ch. 1 Part 15. LBT or synchronisation is implemented in most European countries according to